

03\_CURRENT\_WIRING\_TRUTH

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EVIDENCE PROVENANCE: [PHYSICAL PINOUT]

Direct firmware source code and electrical circuit validation of dual ESP32 prototype boards.

1. Complete Electrical Pinout Table

| Signal Name          | Vehicle A Pin | Vehicle B Pin | Direction | Target Subsystem               |
|----------------------|---------------|---------------|-----------|--------------------------------|
| Left Motor DIR1/2    | GPIO 25, 26   | GPIO 25, 26   | Output    | Left Direction In 1 / 2        |
| Left Motor PWM       | GPIO 27       | GPIO 27       | PWM Out   | LEDC 1 kHz Speed Channel       |
| Right Motor DIR1/2   | GPIO 32, 33   | GPIO 32, 33   | Output    | Right Direction In 1 / 2       |
| Right Motor PWM      | GPIO 14       | GPIO 14       | PWM Out   | LEDC 1 kHz Speed Channel       |
| Motor Standby (STBY) | GPIO 13       | GPIO 13       | Output    | H-Bridge Enable (LOW = High-Z) |
| Wheel Optical Tach   | GPIO 35       | GPIO 35       | Interrupt | LM393 Slotted Disc Detector    |
| IMU I2C (SDA / SCL)  | GPIO 21, 22   | GPIO 21, 22   | I2C Bus   | MPU6050 6-DOF (0x68)           |
| LoRa SPI (SCK/MISO)  | GPIO 18, 19   | GPIO 18, 19   | SPI Bus   | SX1278 433 MHz Module          |
| LoRa SPI (MOSI/SS)   | GPIO 23, 5    | GPIO 23, 5    | SPI Bus   | SX1278 433 MHz Module          |
| LoRa IRQ / Reset     | GPIO 2, 4     | GPIO 2, 4     | Input/Out | DIO0 Packet IRQ / HW Reset     |

2. High-Impedance Standby Protection

On Vehicle B, GPIO 13 directly controls TB6612FNG STBY pin. Holding STBY LOW puts MOSFETs into high-Z mode, physically preventing current to drive motors until explicit software release.